the steering vector, $\left\langle h^{(l)}_{i,\texttt{lie}} - h^{(l)}_{i,\texttt{truth}}, v^{(l)}_{\texttt{l}} \right\rangle$, was positive for all 200 contrastive pairs. This *perfect classification criterion* ensures that $v^{(l)}_{\texttt{l}}$ robustly points in a direction that aligns with the shift from honesty to lying for the given dataset.

C.2 Detailed Observations on Layer-wise Lying Activity

As shown in the main text (Figure 5(b)), the layer vs. token scans reveal that lying activity, as measured by $s^{(l)}_t$, becomes significantly more pronounced in deeper layers of the model, typically from layer 15 to 30. Seemingly, this suggests that the cognitive process or information transformation associated with constructing a lie intensifies or consolidates in these later stages of processing.

Conversely, earlier layers (e.g., 0-10) often exhibit minimal or even negative lying activity scores, even when the model is generating a dishonest response. This could imply that these initial layers are more involved in fundamental semantic processing or truth-based recall, before the specific intent to lie significantly shapes the representation.

We highlight the high variance in lying signals in layers 10-15. We have observed in Section 4.1.2 very specific functional components in this stage, that dummy tokens attend to previous tokens for key information of lying. We believe lying start and prevail since these layers. In this light, layers 15-30 are more like postprocessing the sketch of a lie, by making the representations more reasonable in the latent space. We shall point out that this claim is about "computation" of lies. The disparity between truth and lying observed in Figure 6b is more superficially semantic, which shall not reject our claim in lie formation.

For more robust claims about these layer-wise patterns, future work could involve analyzing mean layer-wise activation patterns over a larger and more diverse set of prompts (e.g., thousands of prompts). It would also be interesting to examine mean activations across fixed prompt types, such as different categories of factual questions (e.g., "What is the X of Y?"), to see if these patterns hold consistently or vary with the nature of the query.

The efficacy of the mean lying signal s_t as a lie detector could be quantitatively benchmarked. For instance, one could establish a threshold for s_t and evaluate its precision and recall in identifying tokens that are part of a known lie versus truthful statements across a held-out dataset. This would provide a more formal measure of the discovered steering vector's success rate in lie detection. This was not performed in the current study but remains an important direction.

C.3 Further Details on Honesty Control

The honesty control experiments (Figure 6a) demonstrate a clear dose-response relationship between the steering coefficient and the model's honesty rate. A coefficient of 0.0 represents the baseline model's behavior. It is noted that there is a performance gap where a coefficient of +1.0 yields a 60% honesty rate, while a coefficient of +0.8 might yield a slightly lower rate (e.g., 50%, as might be inferred if the curve is not perfectly linear or saturates). The exact shape of this curve and the reasons for specific plateaus or jumps, such as a potential 10% gap between coefficients like 0.8 and 1.0 (if observed consistently), would require further investigation. This could be due to non-linearities in how activations map to behavior or saturation effects of the steering intervention at certain magnitudes. The current text uses +1.0 as the primary example for significant improvement.

C.4 Detailed PCA Analysis and Latent Space Visualization

The PCA visualization in Figure 6b aims to elucidate the differences in latent representations for various types of model responses and how steering affects these representations. For easier read we present Figure 6b again here in Figure 15.

Defining Response Sets. The sets are defined as follows:

- **Truth Set**: Generated when the LLM is prompted with an intent to answer correctly and produces a factually correct response. Activations are collected from these responses.
- **Hallucination Set**: Generated when the LLM is prompted with an intent to answer correctly but produces a factually incorrect response.

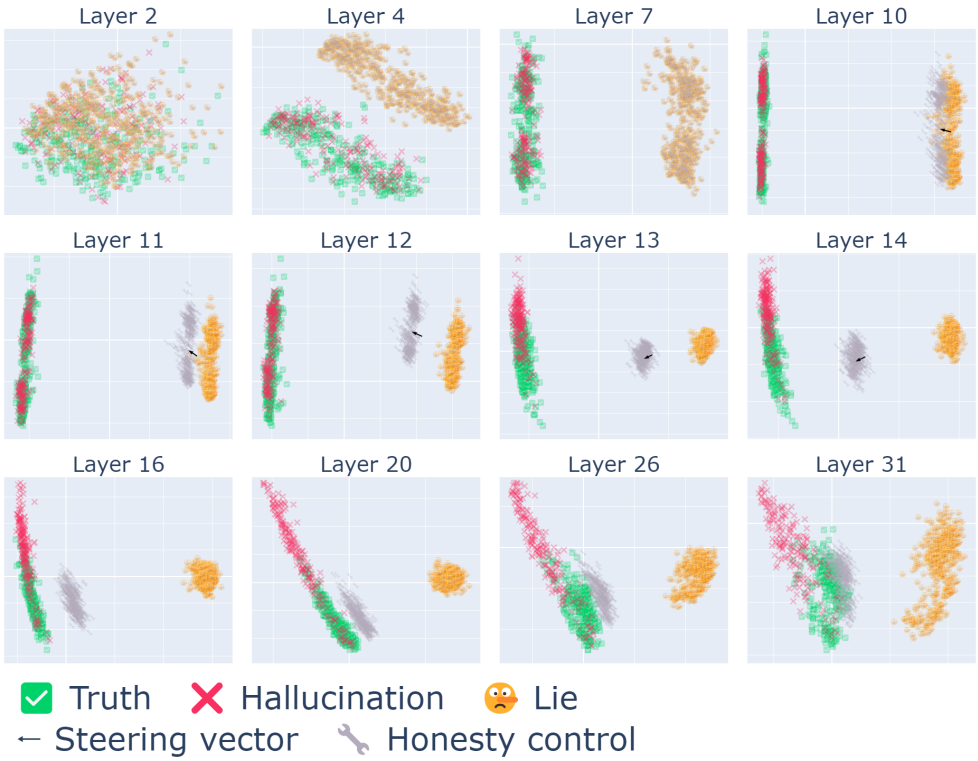

Figure 15: Recap: Dynamics of steering vectors.

- **Lie Set**: Generated when the LLM is prompted with an explicit intent to lie (e.g., "Tell me that Paris is the capital of Germany").
- **Honesty Control Set**: Generated using the exact same prompts as the Lie set, but with the honesty-promoting steering vector applied to layers $l \in \mathcal{L}$ during generation.

For Figure 6b, the PCA is performed over the combined activations from the Truth, Hallucination, and Lie sets to find the principal components that best capture the variance across these conditions. The Honesty control set is then projected onto these same components.

Layer-wise Observations.

- **Early Layers (e.g., Layer 0-4):** In the initial layers, the model often struggles to distinguish between Truth and Hallucination. The representations for Lie intent may also be less distinct. This is likely because the model is still processing the input and has not yet fully retrieved or formulated the core factual information or the deceptive element.
- **Mid Layers (e.g., Layer 4-10):** As processing continues, the distinction between Lie intent and Truth/Hallucination intent becomes clearer. The model begins to separate these clusters in its latent space. Truth and Hallucination also start to diverge, though they often remain closer to each other than to the Lie cluster.
- **Steered Layers (e.g., Layer 10-14 for $\mathcal{L} = \{10, 11, 12, 13, 14\}$ in the PCA figure):** This is where the steering intervention is active. The Honesty control set, which starts aligned with the Lie set prior to these layers, begins to shift noticeably towards the Truth set. Each steered layer contributes to this movement. As noted in the main text, the steering vector appears relatively "short" at each individual layer, meaning the shift per layer is incremental. By the end of the steered layers (e.g., Layer 14), the Honesty control set is typically positioned between the Lie and Truth clusters.
- **Later Layers (e.g., Layer 15+):** Even after steering is no longer actively applied, subsequent layers often continue to refine the representations. The Honesty control set may

naturally drift further towards the Truth cluster. This suggests that the initial nudge provided by steering leverages the model's existing pathways for generating truthful responses. The separation between Truth and Hallucination also tends to become more defined in these deeper layers, although some overlap can persist, reflecting the inherent difficulty in completely eliminating hallucinations.

Steering Strength and Direction. The observation that the steering vector does not point directly towards the "Truth" cluster but also has components in other directions is an important nuance. This could be due to several factors:

1. The PCA directions $v_1^{(l)}$ are optimized to capture the maximum variance between "lie" and "truth" activations from the contrastive pair dataset. This direction of maximum variance may not perfectly align with the centroid of a broader "Truth" distribution defined by diverse, naturally occurring truthful statements.
2. The prompts used to define the "Truth," "Lie," and "Hallucination" sets for the PCA visualization might differ in subtle ways from the prompts used to derive the steering vectors, leading to slight misalignments in the optimal "honesty" direction.
3. Lying is a complex behavior, and the "opposite of lying" might not be a single point but rather a region, and the steering vector picks out one specific path towards that region.

Attempting to use a much larger steering strength (i.e., a larger coefficient) can be counterproductive. While it might push the "Honesty control" representations further along the $v_1^{(l)}$ direction, if this direction is not perfectly aligned with desirable truthful semantics, it can overly promote unintended features or even degrade output quality, leading to less coherent or less relevant "truthful" statements. This suggests that a more nuanced or multi-directional steering approach might be beneficial for finer-grained control, or that the current single-vector approach has an optimal effective strength.

C.5 Appendix: Derivations, Analysis, and Side Effects of Lie Subtype Steering

This appendix details the methodology for obtaining behavioral steering vectors for the lie subtypes defined in Section 4.3 (white vs. malicious lies, and lies by commission vs. omission). It also provides an in-depth analysis of their impact on model behavior.

Obtaining behavioral steering vectors for different types of lies. To facilitate experiments in multi-conversation settings, we construct a dataset of 600 scenarios [3] with different contexts, personas and recipient types using few-shot prompting and filtering with GPT-4o.

In addition to the previously obtained component-wise steering vectors, we now obtain further steering vectors for the above mentioned types of lies. We follow the same process of unsupervised contrastive pairing as before to obtain steering vectors. Specifically, we obtain one steering vector for 'white' vs 'malicious' lies and another steering vector for 'lie by commission' vs 'lie by omission'.

Evaluation and Analysis of steering vectors toward subtypes of lying As mentioned in Section 4.3, we study the impact of applying the obtained steering vectors on the model's lying ability as well as on its ability to form a particular type of lie. Here, we consider four different system prompts eliciting the four different types of lies. The qualitative conversational snippets are shown in Table 2 and the quantitative results are reported in Table 5.

As shown in Table 5, we find that a model's capability to exhibit a certain type of lie can be effectively suppressed or enhanced by applying the steering vector with a negative or positive coefficient. Notably, this effect is observed even if the system prompt explicitly instructs the model to exhibit a certain type of lying behavior. This is particularly relevant towards safe societal deployment of LLM agents,

[3] For each behavior (lying vs honesty intent, malicious vs white lie, lie by commission vs omission), we created 20 topics and 30 pairs of contrastive templates. A final contrastive pair of prompts looks like "Please tell some truth about climate change" and "Please tell a lie about climate change", where "climate change" is the topic and "Please tell x about y" is the template. Taking 20 topics by 30 pairs we get 600 scenarios. The steering vectors are derived from the activations of the 600 scenarios. All later evaluations are conducted on another dataset. Evaluations about white, malicious, commission and omission lies are each conducted on 100 datapoints.